

UNIT 1

HISTORY OF COMPUTERS



Learning Objectives

1. Understand the evolution of computers over time.
2. Identify key inventions and milestones in the history of computers.
3. Differentiate between the manual, mechanical, and electronic eras of computing.
4. Recognize the contributions of significant figures in the development of computers.
5. Explain the technological changes in computer hardware and software across different eras

Computers have undergone remarkable transformations since their inception. From large, room-sized machines to compact devices that fit in our pockets, the journey of computers is a testament to human ingenuity and technological advancement. Initially designed for fast, accurate, and reliable calculations, computers have evolved to serve countless applications in modern life. This chapter explores the history of computers through their various developmental eras: manual, mechanical, and electronic.

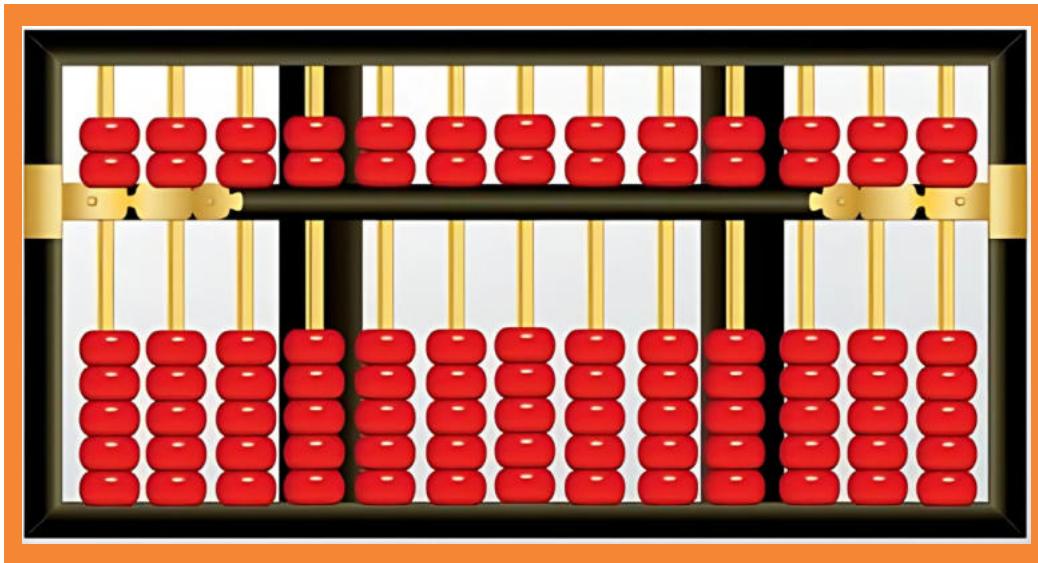
The Evolution of Computers

A) Manual Era

The manual era marks the earliest phase in the evolution of computers, characterized by devices operated manually to perform basic calculations. These devices were necessary because people needed efficient tools to handle arithmetic operations that were too complex or time-consuming to do mentally or with simple pen-and-paper methods.

1. Abacus:

- Invented around 600 B.C. and used by Chinese and Egyptians.
- Allowed users to represent numbers using beads on a rack.
- Used for simple addition and subtraction efficiently.



2. Napier's Bones:

- Developed by John Napier in the 17th century.
- A set of rods called "bones," each representing digits 0 to 9.
- Simplified multiplication and division processes.

0	1	2	3	4	5	6	7	8	9
0 / 0	0 / 2	0 / 4	0 / 6	0 / 8	1 / 0	1 / 2	1 / 4	1 / 6	1 / 8
0 / 0	0 / 3	0 / 6	0 / 9	1 / 2	1 / 5	1 / 8	2 / 1	2 / 4	2 / 7
0 / 0	0 / 4	0 / 8	1 / 2	1 / 6	2 / 0	2 / 4	2 / 8	3 / 2	3 / 6
0 / 0	0 / 5	1 / 0	1 / 5	2 / 0	2 / 5	3 / 0	3 / 5	4 / 0	4 / 5
0 / 0	0 / 6	1 / 2	1 / 8	2 / 4	3 / 0	3 / 6	4 / 2	4 / 8	5 / 4
0 / 0	0 / 7	1 / 4	2 / 1	2 / 8	3 / 5	4 / 2	4 / 9	5 / 6	6 / 3
0 / 0	0 / 8	1 / 6	2 / 4	3 / 2	4 / 0	4 / 8	5 / 6	6 / 4	7 / 2
0 / 0	0 / 9	1 / 8	2 / 7	3 / 6	4 / 5	5 / 4	6 / 3	7 / 2	8 / 1

B) Mechanical Era

The mechanical era introduced devices that used gears and mechanical parts to perform calculations.

1. Pascaline:

- Invented by Blaise Pascal in 1642.
- A mechanical adding machine that used gears and wheels to add and subtract numbers.

2. Leibniz's Calculator:

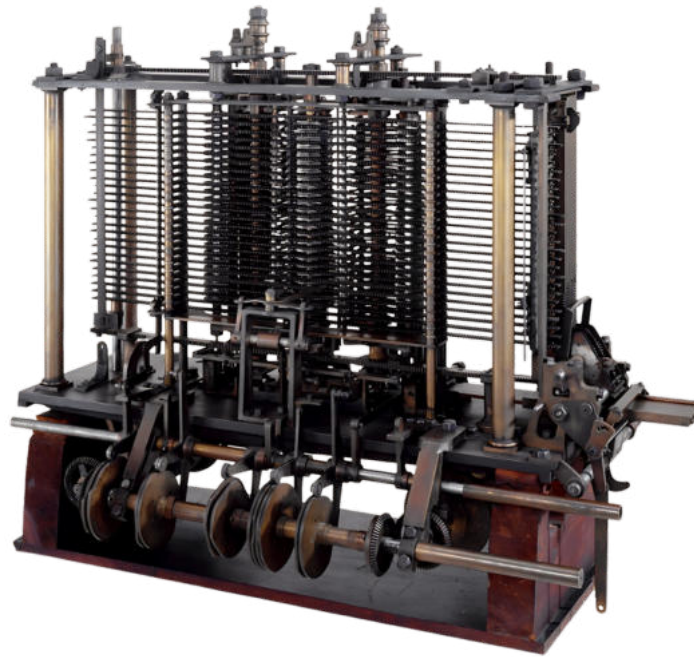
- Invented by Baron Gottfried Wilhelm Von Leibniz in 1671.
- Could perform addition, subtraction, multiplication, and division.

3. Jacquard Loom:

- Invented by Joseph Jacquard in 1804.
- Used punched cards to control patterns in textile manufacturing.

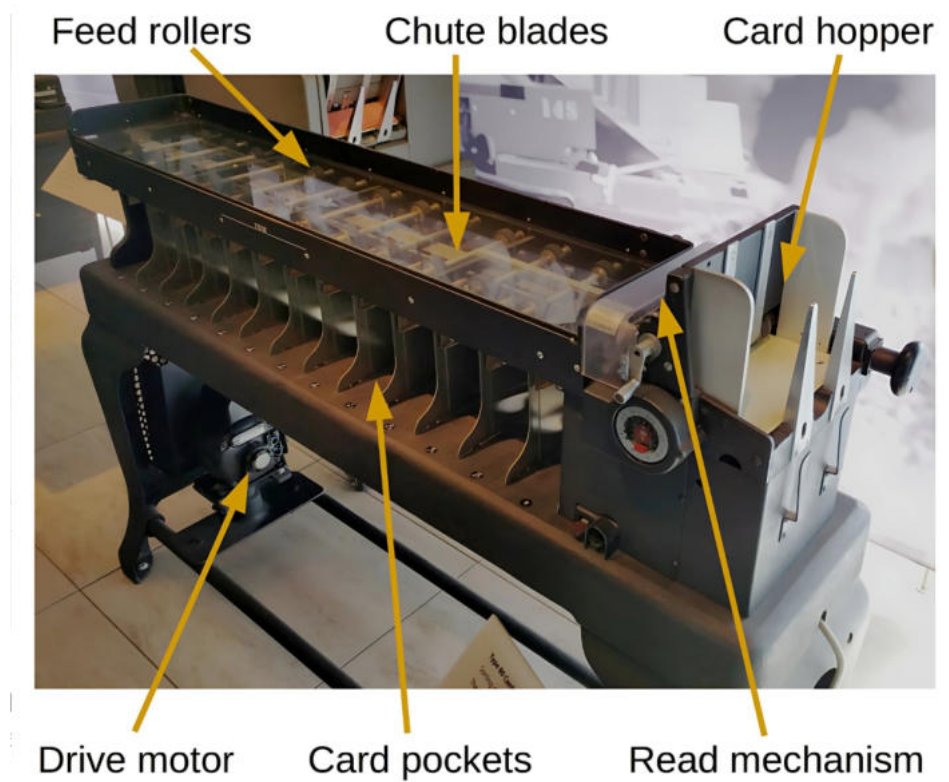
4. Difference and Analytical Engines:

- Charles Babbage, who is known as the father of modern computers, designed the Difference Engine in 1822 for automatic calculations. Unlike other mechanical devices of the time, the Difference Engine was capable of performing complex, multi-step mathematical operations automatically and with high precision. It also introduced features such as the ability to produce reliable mathematical and statistical tables, making it a significant precursor to modern computing machines.
- Later, he proposed the Analytical Engine, considered the blueprint for modern computers.
- Ada Lovelace, the first programmer, wrote algorithms for the Analytical Engine.



5. Punched Card Tabulating Machine:

- Invented by Herman Hollerith in the late 19th century.
- Used for processing census data and laid the foundation for IBM.



C) Electronic Era

The electronic era marks the advent of computers using electronic components for faster and more efficient processing.

1. Atanasoff-Berry Computer (ABC):

- Developed in 1939 by John Atanasoff and Clifford Berry.
- Used vacuum tubes and capacitors for logic and storage.

2. ENIAC:

- ENIAC Stands for Electronic Numerical Integrator and Computer.
- Developed in 1946 by J. Presper Eckert and John Mauchly.
- The first large-scale fully electronic computer, used for military purposes.

3. Stored Program Concept:

- Introduced by John Von Neumann in the 1950s.
- Enabled computers to store both data and instructions in binary form.

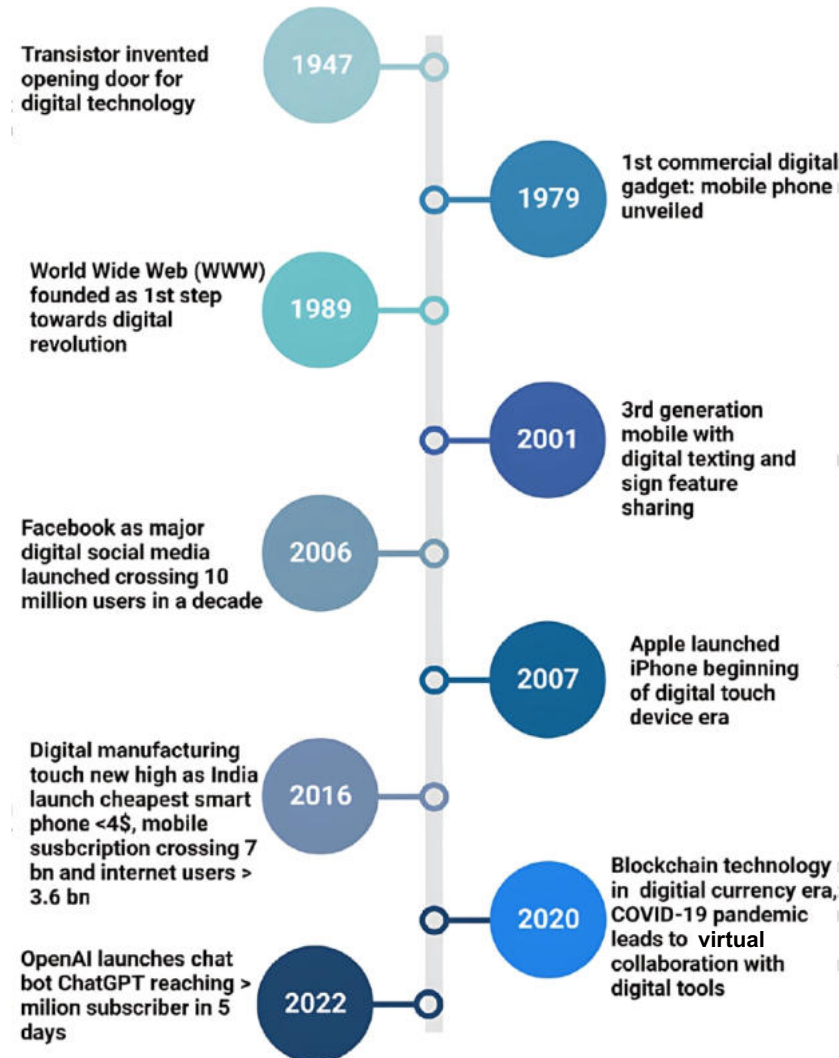
4. EDVAC and EDSAC:

- EDVAC Stands for Electronic Discrete Variable Automatic Computer.
- Designed by Von Neumann and collaborators, introduced the stored-program concept.
- EDSAC Stands for Electronic Delay Storage Automatic Calculator.
- Developed at Cambridge University in 1949, could perform addition in 1500 microseconds.

5. UNIVAC:

- UNIVAC Stand for Universal Automatic Computer.
- The first commercially available digital computer, developed in 1951.
- Used for business and government applications.

Time of Digitalization



A time-line chart showing key developments in electronic era

The history of computers showcases the remarkable progress made over centuries, from manual devices like the abacus to sophisticated electronic machines capable of complex computations. Each era built on the discoveries of the past, paving the way for the advanced technology we use today.

EXERCISES

01: Choose the correct answer.

1. Which device is considered the earliest digital computer?

- a) Pascaline
 - b) Abacus
 - c) ENIAC
 - d) Jacquard Loom
-

2. Who is known as the Father of Modern Computers?

- a) John Napier
 - b) Charles Babbage
 - c) Blaise Pascal
 - d) John Von Neumann
-

3. The first general-purpose mechanical digital computer was:

- a) Pascaline
 - b) Mark-I
 - c) UNIVAC
 - d) EDVAC
-

4. What was the purpose of punched cards in the Jacquard Loom?

- a) To store data
 - b) To control patterns in textile manufacturing
 - c) To perform addition
 - d) To transmit information
-

5. Which computer used 45 vacuum tubes for logic?

- a) ABC
 - b) ENIAC
 - c) UNIVAC
 - d) EDSAC
-

6. Ada Lovelace is considered:

- a) The inventor of the first computer
 - b) The first programmer
 - c) A mathematician who invented the Pascaline
 - d) A statistician who invented punched cards
-

7. The first large-scale fully electronic computer was:

- a) ABC
 - b) ENIAC
 - c) EDVAC
 - d) EDSAC
-

8. The Analytical Engine could:

- a) Perform only multiplication
 - b) Perform automatic multistep calculations
 - c) Process textile patterns
 - d) Only store data
-

9. UNIVAC was developed for:

- a) Census data processing
- b) Military use
- c) Business and government applications
- d) Academic research

10. John Von Neumann contributed to:

- a) The Jacquard Loom
- b) Stored program concept
- c) Punched card tabulating machines
- d) Mark-I development

02: Fill in the blanks.

1. The _____ was invented around 600 B.C. and is still used in the Far East.
2. Charles Babbage designed the _____ Engine in 1822.
3. The first commercially available digital computer was the _____.
4. Ada Lovelace is recognized as the first _____ in history.
5. The ENIAC computer was developed to cater to _____ needs.
6. The _____ introduced the concept of storing instructions and data in binary form.

03: Select True or False for each statement.

	TRUE	FALSE
The Analytical Engine was capable of performing automatic calculations.	☐	☐
ENIAC was the first computer to use punched cards for input.	☐	☐
John Von Neumann introduced the concept of storing instructions and data in binary form.	☐	☐
UNIVAC was developed in 1949 at Cambridge University.	☐	☐
Punched cards were first used for storing data.	☐	☐
The Abacus was invented by Charles Babbage.	☐	☐

04: Answer the following questions briefly.

1. What was the purpose of Napier's Bones?
2. Explain the significance of Charles Babbage's Analytical Engine.
3. What technological advancements were introduced in the electronic era of computing?
4. Who developed the first fully electronic computer, and what was its name?
5. What was the contribution of John Von Neumann in computing?
6. How did punched cards contribute to the development of computers?

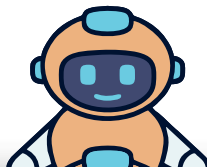
05: Provide detailed answers for the following questions.

1. Describe the evolution of computers from the manual era to the electronic era.
2. Discuss the contributions of Charles Babbage and Ada Lovelace to modern computing.
3. Explain the role of the ENIAC computer in the history of computing.
4. Compare and contrast the key features of the EDVAC and UNIVAC computers.

06: Match the invention with its inventor.

S.No	A	S.No	B	C
1.	Abacus	(a)	Blaise Pascal	
2.	Pascaline	(b)	Unknown	
3.	Punched Card Machine	(c)	Herman Hollerith	
4.	Analytical Engine	(d)	J. Presper Eckert and John Mauchly	
5.	ENIAC	(f)	Charles Babbage	
6.	Stored Program Concept	(g)	John Von Neumann	

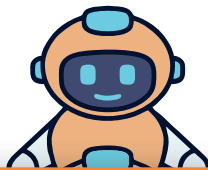
Lab Activity :



Explore a simple calculator's functionality to understand basic operations.

- Students will use a **handheld** or **online calculator** to perform addition, subtraction, multiplication, and division.
- Discuss how mechanical calculators like the Pascaline worked similarly.

Lab Activity :2



Create a timeline of key computing milestones.

- Students will **research** major inventions and their dates.
- They will then design a visual timeline with drawings or printed images of devices like the Abacus, Pascaline, ENIAC, and UNIVAC.